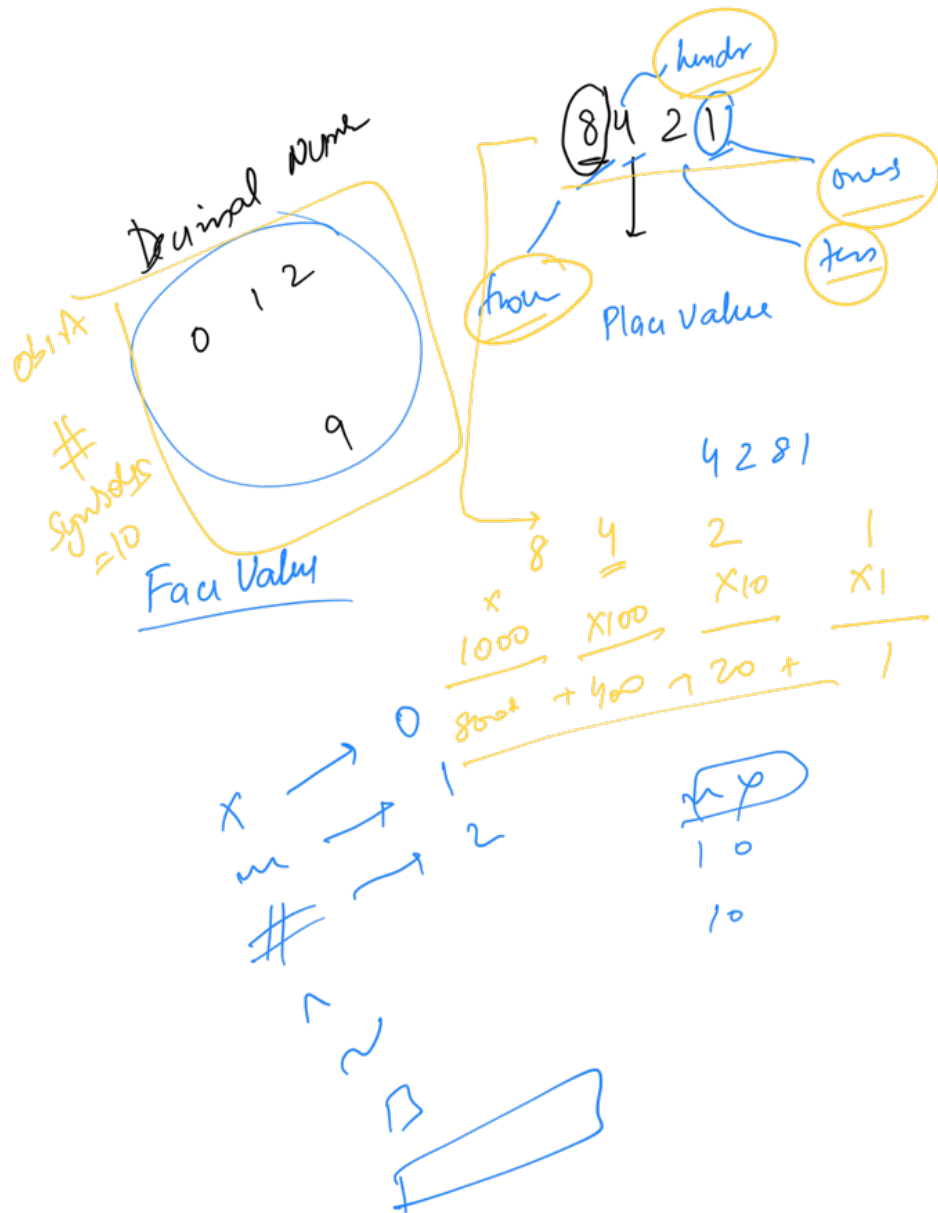


Bit Manipulation

Number System



$$\text{Value} = \sum_{\text{digit}} \text{face value} \times \text{place value}$$



Decimal
Number
System
place
value



obs
B

place value inc
by a power of 10

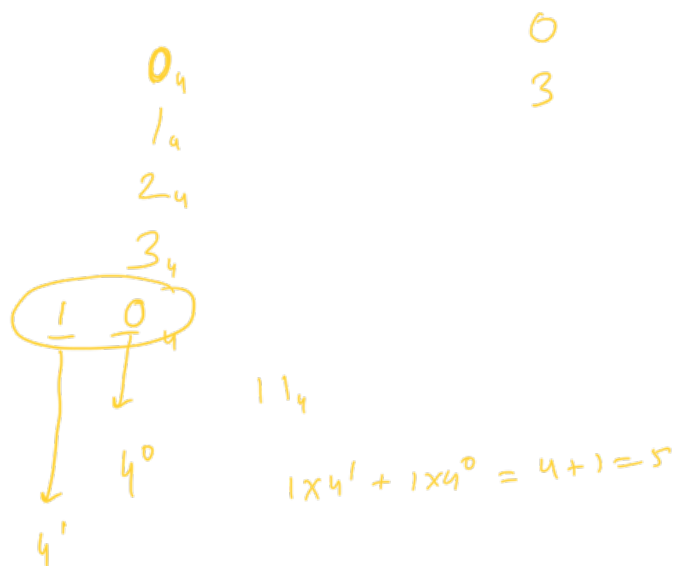
pv (i^{th} digit from the right)

$$= 10^{i-1}$$

$$i = 1, 2, 3, 4, 5, 6, 7, 8$$

obs A : 10 symbols

obs B : Place value inc by a power of 10



$$1 \times 4^1 + 0 \times 4^0$$

$$= 4 + 0$$

$$= 4$$



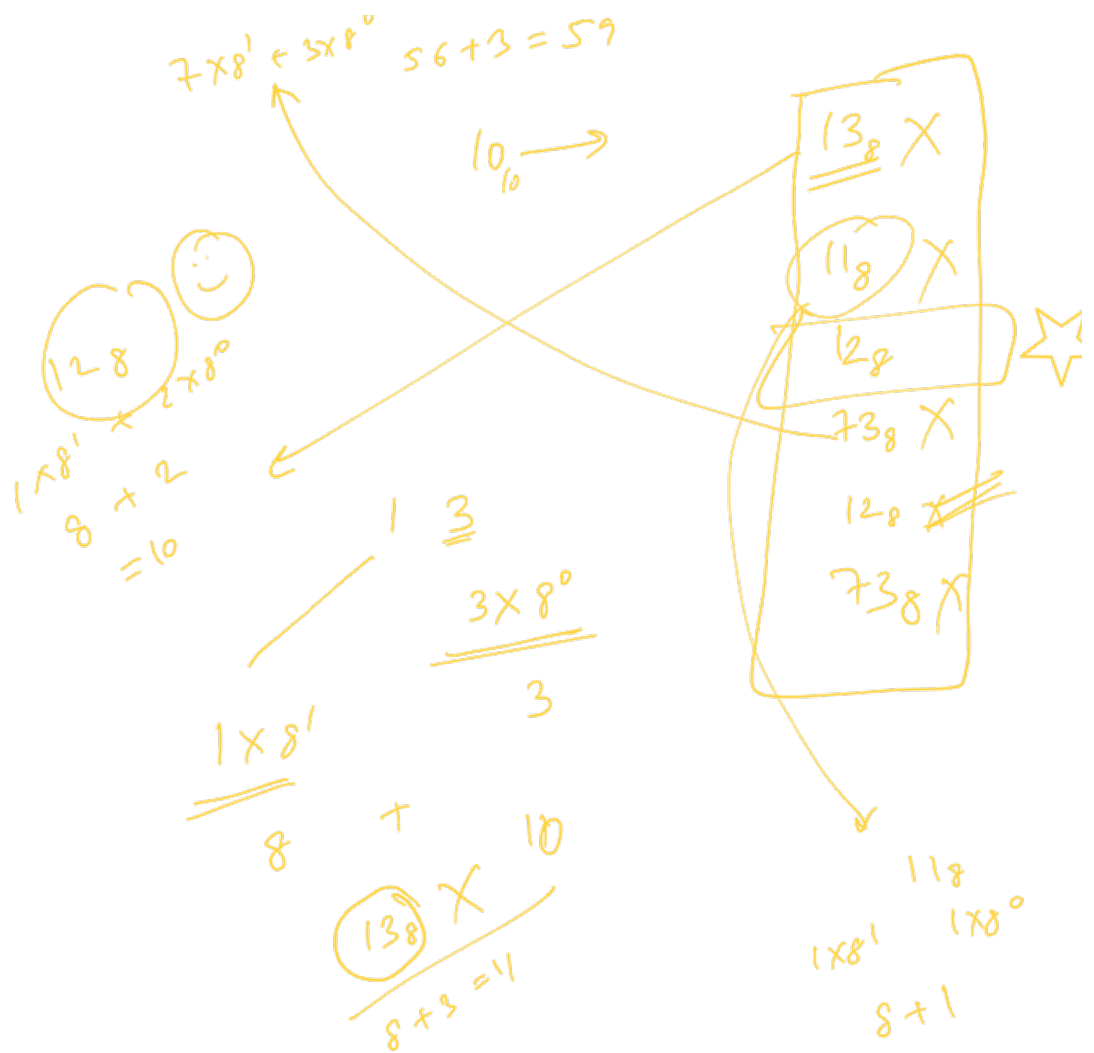
$$\begin{array}{c} \textcircled{0} \\ \hline B^3 \quad B^2 \quad B^1 \quad B^0 \end{array}$$

$$\underline{\underline{(B)^2}} \times 0$$



Octal [8 symbols]

0 1 2 3 4 5 6 7



decimal conversion (with a smiley face)

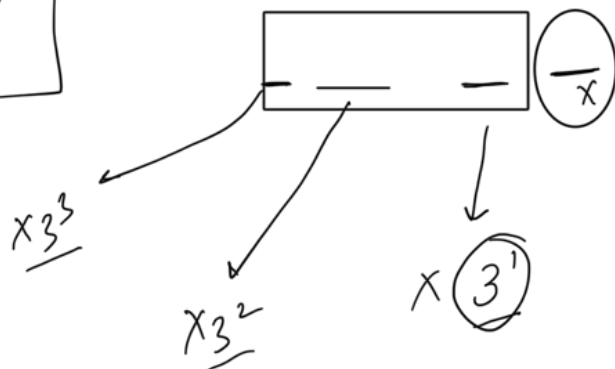
Any base to decimal conversion

$$\begin{aligned}
 & \underline{(471)_9} \\
 & (4 \times 9^2) + (7 \times 9^1) + (1 \times 9^0) \\
 & \underline{324} + 63 + 1 \\
 & \underline{388}_{10} \star
 \end{aligned}$$

Decimal to Any base B

$$\begin{array}{ccc}
 42_{10} & \longrightarrow & 10101_3 \\
 \text{Decimal} & & \text{Ternary}
 \end{array}$$

Symbol	
0	✓
1	✓
2	✓



1

714 divider

$$\begin{array}{r}
 \textcircled{4} \\
 3 \overline{) 14} \\
 \underline{-12} \\
 \textcircled{2}
 \end{array}
 \qquad
 \begin{array}{r}
 \textcircled{42} \\
 3 \overline{) 42} \\
 \underline{-3} \\
 12 \\
 \underline{-12} \\
 \textcircled{0} \text{ remainder}
 \end{array}$$

$$\begin{array}{r}
 \textcircled{0} \\
 3 \overline{) 1} \\
 \underline{-0} \\
 1
 \end{array}
 \qquad
 \begin{array}{r}
 \textcircled{1} \\
 3 \overline{) 4} \\
 \underline{-3} \\
 1
 \end{array}
 \qquad
 \begin{array}{r}
 1 \ 2 \ 0 \\
 \underline{3}
 \end{array}$$

$$42_{10} \equiv (1120)_3 \quad \textcircled{\begin{smallmatrix} \cdot \\ 0 \end{smallmatrix}}$$

$$1 \times 3^3 + 1 \times 3^2 + 2 \times 3^1 + 0 \times 3^0$$

$$27 + 9 + 6 + 0$$

$$36 + 6 = \textcircled{42}$$

$$\underline{90}_{10} \equiv \underline{\underline{(??)}}_8$$

$$\begin{array}{r}
 11 \\
 8 \overline{) 90} \\
 \underline{-8} \\
 10 \\
 \underline{-8} \\
 2
 \end{array}$$

Our

$$(02101)_3$$

$$\begin{array}{r} 1 \times 3^2 \\ + 0 \times 3^1 \\ + 1 \times 3^0 \\ \hline 9 \\ 0 \\ 1 \\ \hline 10 \end{array}$$

9

54

64



Base 11

$$A \equiv 10$$

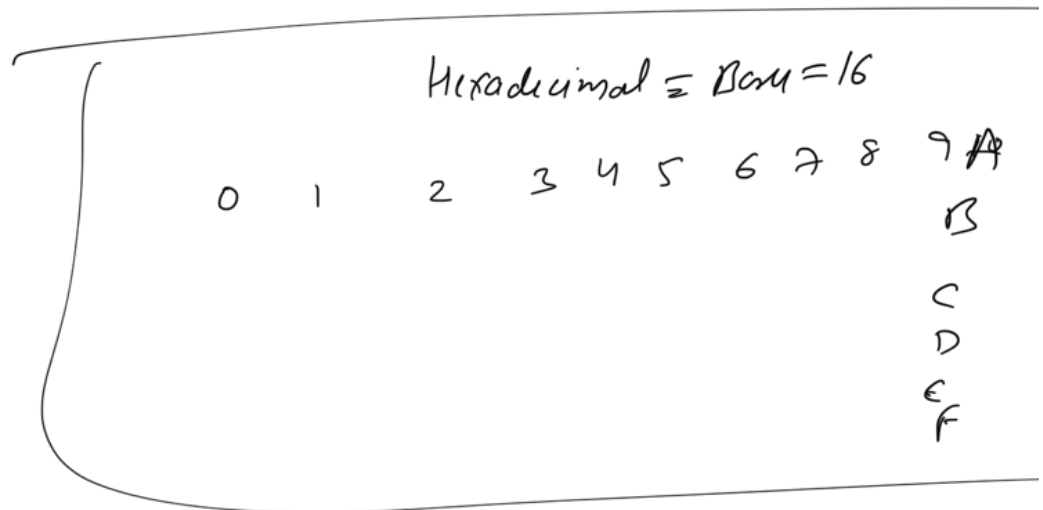
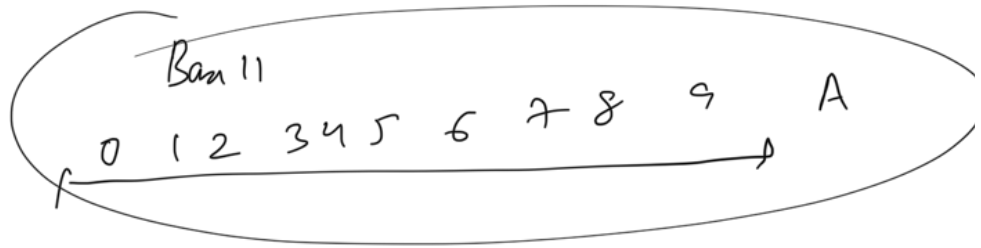
$$B \equiv 11$$

$$C \equiv 12$$

$$D \equiv 13$$

$$E \equiv 14$$

$$F \equiv 15$$



$$(AC1)_{16}$$

$$1 = 1 \times 16^0$$

$$192 = 12 \times 16^1$$

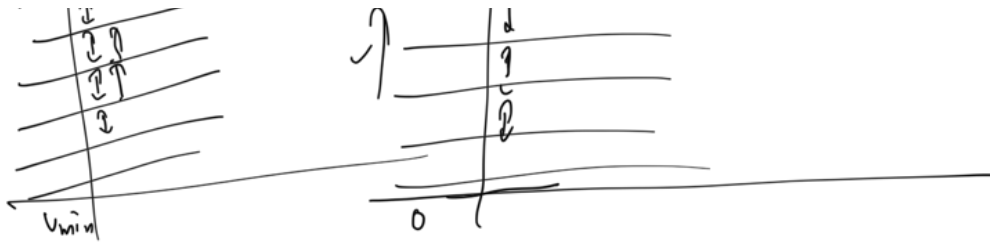
$$2560 = 10 \times 16^2$$

$$\underline{\underline{2753}}_{10}$$

Decimal = 10

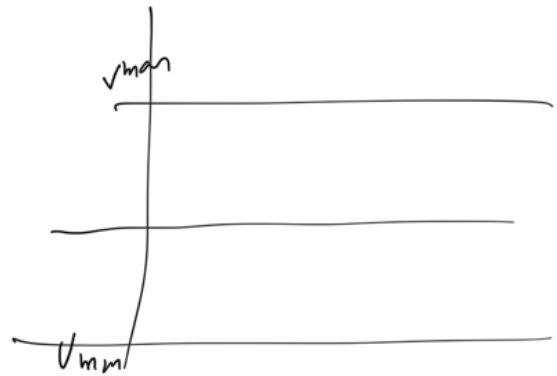
~~Value~~

240	0 $\equiv$ 9
	0 $\equiv$ 8



Binary
Number System

2

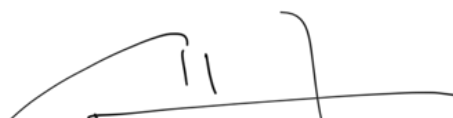


1 1 0 1 0 1 0 0 1

↑
high
✓
≡
ON

↓
low
✓
≡
OFF

$$87_{10} = (?)_2$$



$$\begin{array}{r}
 22 \\
 -22 \\
 \hline
 0
 \end{array}$$

$$\begin{array}{r}
 2 \\
 \hline
 2 \\
 -2 \\
 \hline
 0
 \end{array}$$

$$\begin{array}{r}
 5 \\
 \hline
 11 \\
 -10 \\
 \hline
 1
 \end{array}$$

$$\begin{array}{r}
 5 \\
 \hline
 5 \\
 -4 \\
 \hline
 1
 \end{array}$$

$$\begin{array}{r}
 0 \\
 \hline
 1 \\
 -0 \\
 \hline
 1
 \end{array}$$

$(10110)_2$

Addition

Decimal

$$\begin{array}{r}
 1717 \\
 9815 \\
 \hline
 1532
 \end{array}$$

$$150/10 = 5$$

$$12/10 = 2$$

$$\begin{array}{r}
 10 \overline{) 10} \\
 \underline{10} \\
 0
 \end{array}$$

$$\begin{array}{r}
 11 \\
 3459 \\
 \underline{2897} \\
 6306
 \end{array}$$

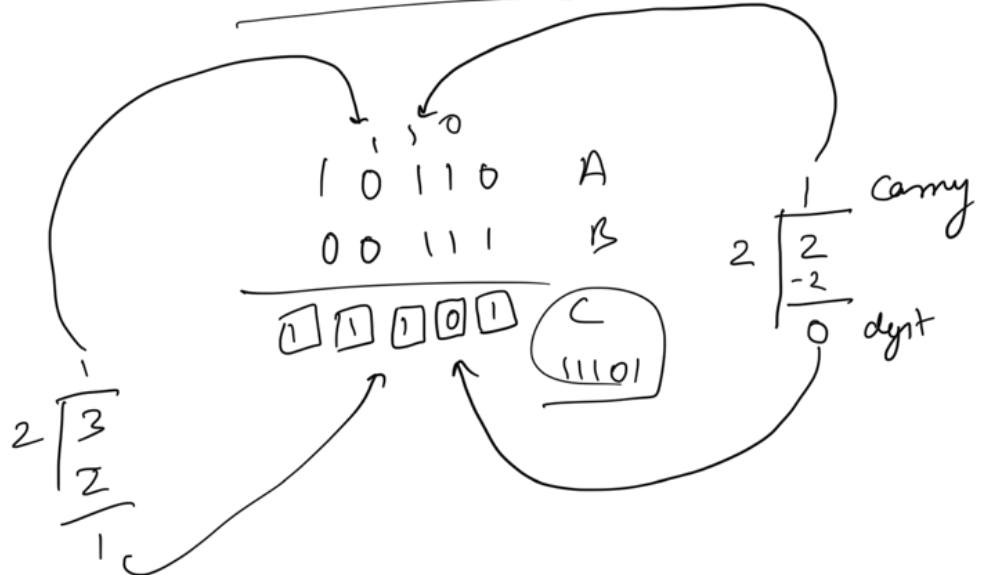
$$\begin{array}{r}
 10 \overline{) 16} \\
 \underline{10} \\
 6
 \end{array}$$

quotient
carry

digit
remainder

↑

Addition in Binary



8:47
8:49

Bit Manipulation

$[0, 1, 2, \dots, 9] \equiv$

10 Symbols of Decimal Number System

Symbols
111

$\equiv$ Digit

111

$[0, 1] \equiv$

2 Symbols of Binary Number System

$\equiv$

Bit

U-1-

U sym

Bitwise Operations

✓ $\&$ Bitwise AND

✓ $|$ Bitwise OR

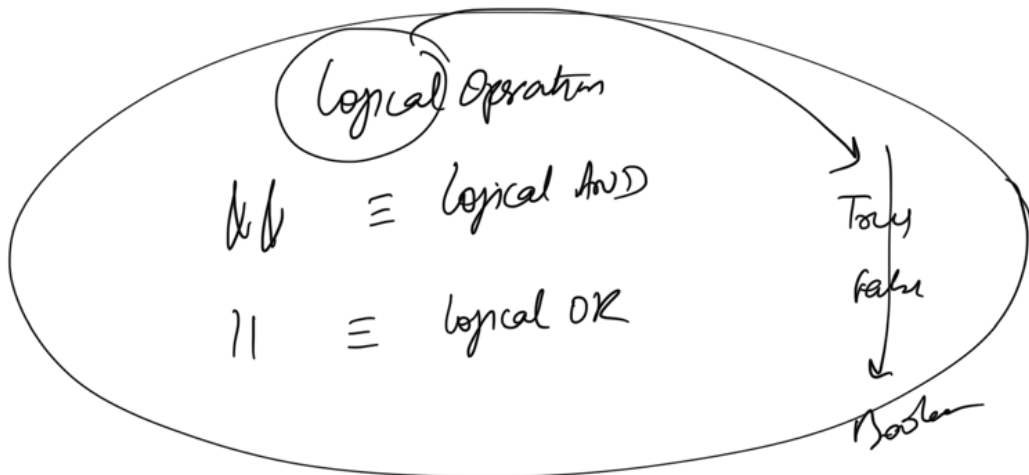
✓ $\sim$ Bitwise NOT $\equiv$ unad on 1 operand

✓ $\wedge$ Bitwise XOR

$\ll$
 $\gg$ Not today

$10 \oplus 20 =$
 $\oplus$
 $\otimes$
 $\oplus$

Logical Opr
True $\&$ True = T
True $\&$ False = F



Bitwise AND

	0	1	0	0	0	1	1
k	0	1	0	0	1	1	1
	0	1	0	0	0	1	1

if	answer
0 0	0
0 1	0
1 0	0
1 1	1

$$\begin{array}{r}
 111011 \\
 110010 \\
 \hline
 110010
 \end{array}$$

Bitwise OR

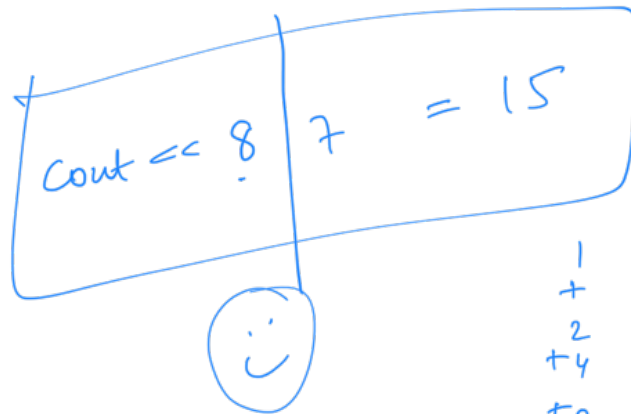
$$\begin{array}{r}
 111011 \\
 110010 \\
 \hline
 111011
 \end{array}$$

	1	
0		0
0		1
1		0
1		1



111011

if atleast 1 of
the 2 inputs
was 1



8 $\equiv$ 1000
7 $\equiv$ 0111
= 1111

1
+
2
+4
+8

15

$a = 13$

$b = 10 \equiv$

1101
1010

1000

$a \& b$
= 8

$$\text{print}(13 \& 10) = 8$$

$$\begin{array}{r} 10 \equiv 1010 \\ 1 \equiv 0001 \\ \hline 1011 \\ 8 \& 1 = 11 \end{array}$$

$$\sim (10100)_2$$

$$\begin{array}{r} \sim 10100 \\ \hline 01011 \end{array}$$

$$\sim 4$$

$$\begin{array}{r} \sim 100 \\ \hline 011 = 3 \end{array}$$

$$\sim 4 = 3$$

Bitwise XOR

exclusive OR

^

$$\begin{array}{rcl}
 0 \wedge 0 & = & 0 \\
 0 \wedge 1 & = & 1 \\
 1 \wedge 0 & = & 1 \\
 1 \wedge 1 & = & 0
 \end{array}$$

XOR of 2 inputs
 = HIGH

$$A = 15 \equiv$$

1111

$$B = 6 \equiv$$

0110

$$\begin{array}{rcl}
 \text{print}(A \& B) & = & 0110 = 6 \\
 \text{print}(A | B) & = & 1111 = 15 \\
 \text{print}(A \wedge B) & = & 1001 = 9
 \end{array}$$



Quick way of convert
 D to B

$$A = 20$$

$$\begin{array}{r}
 20 \\
 - 16 \\
 \hline
 4 - 4 = 0
 \end{array}$$

$$\frac{1}{2^4} \quad \frac{0}{2^3} \quad \frac{1}{2^2} \quad \frac{0}{2^1} \quad \frac{0}{2^0}$$

$$A = 10$$

$$10 - 16 = 2 - 0 =$$

$$A = 10$$

$$10 - 10 - - - -$$

$$\underline{1} \quad \underline{0} \quad \underline{0} \quad \underline{1} \quad \underline{0}$$

$$A = 250$$

$$2^7 = 128$$

$$\begin{array}{r} 250 \\ -128 \\ \hline 122 \\ -64 \\ \hline 58 \\ -32 \\ \hline 26 \\ -16 \\ \hline 10 \\ -8 \\ \hline 2 \\ -2 \\ \hline 0 \end{array}$$

$$250_{10} \equiv \underline{1} \underline{1} \underline{1} \underline{1} \underline{1} \underline{0} \underline{1} \underline{0}$$

$$7 \quad 6 \quad 5 \quad 4 \quad 3 \quad 2 \quad 1 \quad 0$$

$$A) \boxed{a \mid a = a}$$

$$\begin{array}{r} 1 \mid 010111 \\ 1 \mid 1010111 \end{array}$$

$$5 \begin{smallmatrix} 1 \\ 5 \end{smallmatrix} \equiv \begin{pmatrix} 101 \\ 101 \\ \hline 101 \end{pmatrix} \quad \begin{array}{r} 101 \\ 101 \\ \hline 101 \end{array}$$

$$\textcircled{B} \quad a \& a = a$$

$$\textcircled{C} \quad a \wedge a = 0$$

$$\begin{array}{r} 101 \\ \wedge 101 \\ \hline \end{array}$$

$$\begin{array}{l} \checkmark \\ \checkmark \\ \checkmark \\ \checkmark \end{array} \quad \begin{array}{l} a \& a = a \\ a | a = a \\ \hline a \wedge a = 0 \end{array}$$

$$\begin{array}{l} a \& 0 = 0 \\ a | 0 = a \\ a \wedge 0 = a \end{array} \quad \begin{array}{l} \checkmark \\ \checkmark \\ \checkmark \\ \checkmark \end{array}$$

$$\begin{array}{l} a \& b == b \& a \\ a | b == b | a \\ a \wedge b == b \wedge a \end{array}$$

Order of operands
do NOT 
matter

$$a \wedge b \wedge c \wedge d = b \wedge c \wedge a \wedge d = a \wedge d \wedge c \wedge b$$

$$a | b | c = b | c | a = c | a | b$$

$$10^A 5^B 10^C = 10^A 10^B 5^C$$

$$\begin{array}{r}
 10 \ 10101 \\
 11 \ 10001 \\
 11 \ 00010 \\
 \hline
 000000
 \end{array}
 \begin{array}{l}
 A \\
 B \\
 C
 \end{array}$$

$$\begin{array}{r}
 A \ 10101 \\
 B \ 1110001 \\
 C \ 1100010 \\
 \hline
 1110111
 \end{array}$$

$$\begin{array}{r}
 A \ 1010101 \\
 B \ 1110001 \\
 C \ 1100010 \\
 \hline
 1000110
 \end{array}$$

For Bitwise AND $\equiv$ all should be 1
 For Bitwise OR $\equiv$ atleast 1 bit should be 1

For Bitwise XOR = odd # of bits should be 1

Q Shree

Combani

Aditya

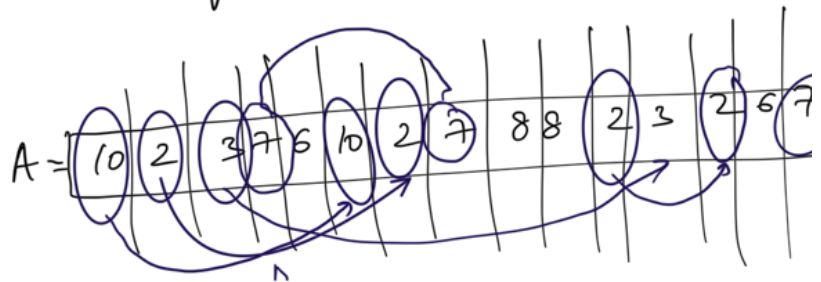
Answer

MS

$$A = \text{of sin } N$$

Where every number is present

Even number of times, but 1 number is present odd no of times



✓ # (10) 2 # (7) 3 **Odd**

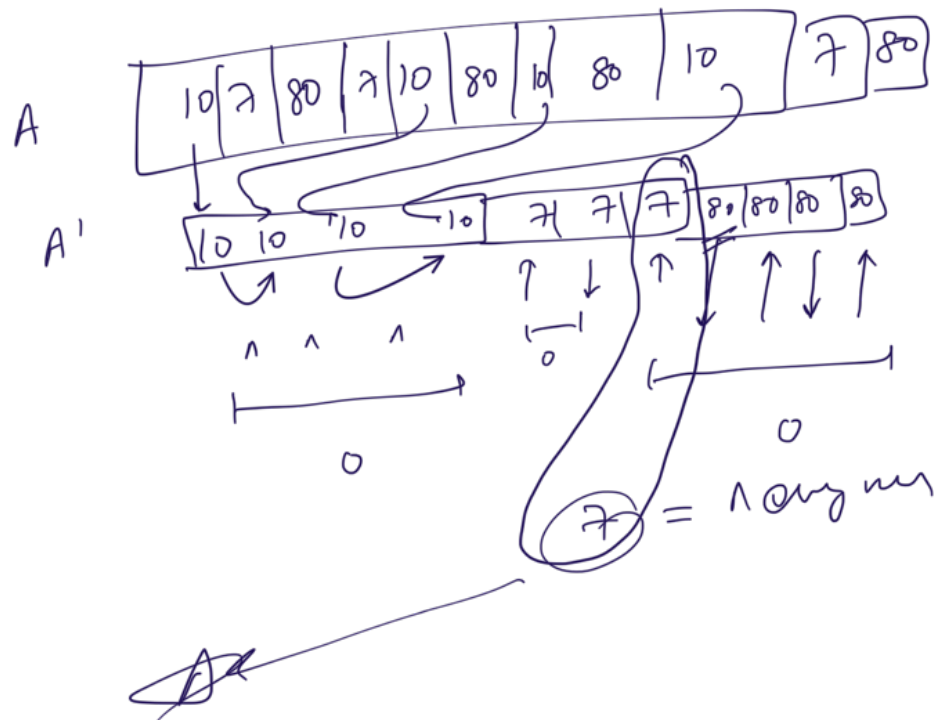
✓ # (2) 4 # (8) 2 ✓

✓ # (3) 2 # (6) 2 ✓

Q Find the number which was present odd number of times

7

$\wedge \equiv$ Excluding



😊
 $T.C = O(N)$
 $S.C = O(1)$
 ans

$ans = a(0)$

$\left[\begin{array}{l} \text{for } (i=1, i < N, i++) \\ \quad ans = ans \wedge a(i) \end{array} \right.$

$\text{print}(ans)$

😊

$\wedge \equiv O(1)$ time

$\Delta \equiv O(1)$

$| \equiv O(1)$

Summary

- ① Number System
- ② Binary No System